

Questions

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Q1 - 2024 (04 Apr Shift 2)

Let $S = \{ \sin^2 2\theta : (\sin^4 \theta + \cos^4 \theta) x^2 + (\sin 2\theta)x + (\sin^6 \theta + \cos^6 \theta) = 0 \text{ has real roots} \}$. If α and β be the smallest and largest elements of the set S , respectively, then $3((\alpha - 2)^2 + (\beta - 1)^2)$ equals _____

Q2 - 2024 (05 Apr Shift 1)

Suppose $\theta \in \left[0, \frac{\pi}{4}\right]$ is a solution of $4 \cos \theta - 3 \sin \theta = 1$. Then $\cos \theta$ is equal to :

(1) $\frac{4}{(3\sqrt{6}+2)}$

(2) $\frac{6+\sqrt{6}}{(3\sqrt{6}+2)}$

(3) $\frac{4}{(3\sqrt{6}-2)}$

(4) $\frac{6-\sqrt{6}}{(3\sqrt{6}-2)}$

Q3 - 2024 (05 Apr Shift 2)

The number of solutions of $\sin^2 x + (2 + 2x - x^2) \sin x - 3(x - 1)^2 = 0$, where $-\pi \leq x \leq \pi$, is _____

Questions

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Answer Key

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Solutions

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Q1

$$D = (\sin 2\theta)^2 - 4 \left(1 - \frac{\sin^2 2\theta}{2}\right) \left(1 - \frac{3}{4} \sin^2 2\theta\right)$$

$$= (\sin 2\theta)^2 - 4 \left(1 - \frac{5}{4} \sin^2 2\theta + \frac{3}{8} \sin^4 2\theta\right)$$

$$D = -\frac{3}{2} \sin^4 2\theta + 6 \sin^2 2\theta - 4 > 0$$

$$3 \sin^4 2\theta - 12 \sin^2 2\theta + 8 < 0$$

$$\sin^2 2\theta = \frac{12 \pm \sqrt{12^2 - 12 \cdot 8}}{6} = \frac{12 \pm 4\sqrt{3}}{6} = \frac{6 \pm 2\sqrt{3}}{3}$$

$$\sin^2 2\theta = 2 \pm \frac{2}{\sqrt{3}}, \text{ but } \sin^2 2\theta \in [0, 1]$$

$$\therefore \alpha = 2 - \frac{2}{\sqrt{3}}, \beta = 1 \rightarrow (\alpha - 2)^2 = \frac{4}{3}, (\beta - 1)^2 = 0$$

$$3(\alpha - 2)^2 + (\beta - 1)^2 = 4$$

Q2

$$4 \left(\frac{1 - \tan^2 \theta/2}{1 + \tan^2 \theta/2} \right) - 3 \left(\frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}} \right) = 1$$

$$\text{let } \tan \frac{\theta}{2} = t$$

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Solutions

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$$\frac{4 - 4t^2 - 6t}{1 + t^2} = 1$$

$$4 - 4t^2 - 6t = 1 + t^2$$

$$\Rightarrow 5t^2 + 6t - 3 = 0$$

$$\Rightarrow t = \frac{-6 \pm \sqrt{36 - 4(5)(-3)}}{2(5)}$$

$$= \frac{-6 \pm \sqrt{96}}{10}$$

$$= \frac{-6 \pm 4\sqrt{6}}{10}$$

$$t = \frac{-3 + 2\sqrt{6}}{5}$$

$$\cos \theta = \frac{1 - t^2}{1 + t^2} = \frac{1 - \left(\frac{2\sqrt{6}-3}{5}\right)^2}{1 + \left(\frac{2\sqrt{6}-3}{5}\right)^2} = \frac{1 - \left(\frac{24+9-12\sqrt{6}}{25}\right)}{1 + \left(\frac{24+9-12\sqrt{6}}{25}\right)}$$

$$= \frac{25 - 33 + 12\sqrt{6}}{25 + 33 - 12\sqrt{6}} = \frac{12\sqrt{6} - 8}{58 - 12\sqrt{6}} = \frac{6\sqrt{6} - 4}{29 - 6\sqrt{6}} \times \frac{29 + 6\sqrt{6}}{29 + 6\sqrt{6}}$$

$$= \frac{100 + 150\sqrt{6}}{625 - 200} = \frac{4 + 6\sqrt{6}}{25 - 8} \times \frac{4 - 6\sqrt{6}}{4 - 6\sqrt{6}}$$

$$= \frac{-200}{25(4 - 6\sqrt{6})} = \frac{-8}{4 - 6\sqrt{6}} = \frac{4}{3\sqrt{6} - 2}$$

Q3

$$\sin^2 x - (x^2 - 2x - 2) \sin x - 3(x - 1)^2 = 0$$

$$\sin^2 x - (x - 1)^2 \sin x - 3(x - 1)^2 = 0$$

roots :

$$\begin{array}{cc} & \swarrow \quad \searrow \\ -3 & \quad (x-1)^2 \end{array}$$

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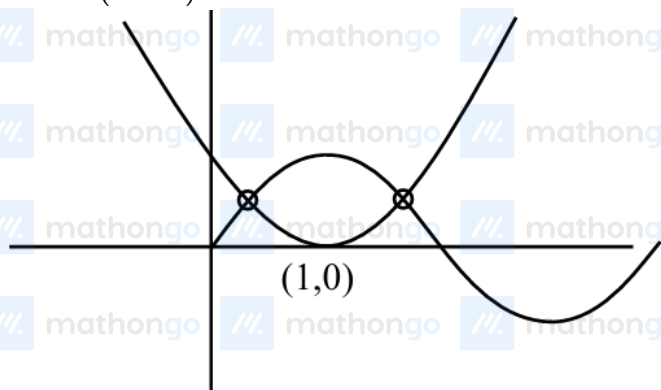
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Solutions

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$$\sin x = -3 \text{ (reject) or } (x-1)^2$$

$$\sin x = (x-1)^2$$



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